INCH-POUND
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DETAIL SPECIFICATION

WINDOWS, NON-ICING, FLAT LAMINATED GLASS, ELECTRICALLY HEATED WITH CONTROLS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers non-icing, electrically heated, flat laminated glass windows contained in metal frames, with temperature controls necessary for operation. The windows are to provide clear vision aboard ships during cold weather operating conditions.

1.2 Classification. Controllers for windows are of the following types, as specified (see 6.2):

a. Type I – Remote controller actuated by a temperature sensor embedded in the window.

b. Type II (for retrofit use only) – Surface controller actuated by an integral temperature sensor for controller replacement on windows not having embedded sensors.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490 - Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

TT-P-1757 - Primer Coating, Alkyd Base, One Component

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-G-2857	-	Glass, Heat-Treated, Glazing, Rectangular (for Bridge Windows)
MIL-PRF-8805	-	Switches and Switch Assemblies, Sensitive, Snap Action (Basic, Limit, Push Button and Toggle Switches), General Specification for
MIL-DTL-15090	-	Enamel, Equipment, Light Gray (Navy Formula No. 111)
MIL-PRF-28750	-	Relays, Solid State, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-1399-300-1	-	Department of Defense Interface Standard, Section 300, Part 1, Low Voltage Electric Power, Alternating Current
MIL-STD-1399-300-2	-	Department of Defense Interface Standard, Section 300, Part 2, Medium Voltage Electric Power, Alternating Current

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

804-1633251	-	Window, Heated, ¾" Panel, Fixed Frame, Assembly & Details (CRES)
804-1633254	-	Window, Heated, 19 X 25, 2"Panel, Fixed Frame, Assembly & Details (CRES)
804-1633257	-	Window, Heated, ½"Panel, Fixed Frame, Assembly & Details (CRES)
804-1633259	-	Window, Heated, ½"Panel, Hinged Frame, Assembly & Details (CRES)
804-1633260	-	Window, Heated, ½"Panel, Hinged Frame, Details (CRES & AL)
804-1633262	-	Window, Heated, ½"Panel, Fixed Frame, Assembly & Details (AL)
804-1633263	-	Window, Heated, ½" Panel, Hinged Frame, Assembly (AL)
804-1633264	-	Window, Heated, ½" Panel, Hinged Frame, Details (AL)
804-2226902	-	Window, Heated, Temperature Controller (Type I) & Glass Details
804-4353964	-	Window, Heated, Temperature Controller (Type II)

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(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

AIA NAS411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM C1036 - Standard Specification for Flat Glass

ASTM C1048 - Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass

ASTM C1172 - Standard Specification for Laminated Architectural Flat Glass

ASTM D1003 - Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics

ASTM D1044 - Standard Test Method for Resistance of Transparent Plastics to Surface Abrasion

(Copies of these documents are available online at www.astm.org.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910.1200 - Hazard Communication

(Copies of this document are available online at www.osha.gov.)

SAE INTERNATIONAL

SAE AS50151 - Connectors, Electrical, Circular Threaded, AN Type, General Specification for

ANSI/SAE Z26.1 - American National Standard for Safety Glazing Materials for Glazing Motor Vehicles and Motor Vehicle Equipment Operating on Land Highways

(Copies of these documents are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Windows. Windows shall be constructed of laminated glass of two or more plies of glass and of glazing qualities in accordance with MIL-G-2857, ASTM C1048, ASTM C1036, ASTM C1172, ASTM D1003, and/or ASTM D1044. The laminate shall be in accordance with ANSI/SAE Z26.1. Glass, plastic, or interlayer plies shall not be notched or drilled post lamination for purposes of inserting sensing or other electrical components. The flat laminated glass panel shall incorporate transparent electrically conductive metallic films for the purpose of heating. Contact with the conductive film shall be maintained through bus bars internal to the laminate that have been laminated within the transparency and are in contact with the film. Windows shall contain two, three, or four temperature sensors (one spare for each circuit) embedded within the interlayer and shall use a type I controller (see 1.2). Four sensors are required for fail-safe controller operation (see 3.4).

3.3.1 Sensor characteristics. Embedded sensors used with type I controllers shall have the following characteristics:

a. Sensor material: Nickel wire.

b. Resistance: 326 ohms ±1 percent at 90 °F.

c. Temperature coefficient: 0.003 ohm/°F nominal at 90 °F (measured at the input to the controller).

d. Size: Approximately 1½inches by 1½inches.

e. Location: In accordance with Drawing 804-2226902.

3.3.2 Frames. The glass shall be contained in frames of corrosion-resistant steel (CRES), bronze, or aluminum as specified (see 6.2). Details shall be in accordance with Drawings 804-1633251, 804-1633254, 804-1633257, 804-1633259, 804-1633260, 804-1633262, 804-1633263, and 804-1633264, as applicable.

3.3.3 Gaskets. The glass shall be fitted in the frame by means of a rubber channel gasket of 30 to 40 durometers. Corners of gaskets shall be vulcanized or otherwise sealed against water penetration. Gaskets of ⅛-inch thickness shall be provided to enable window frames to be installed.

3.3.4 Junction box. The junction box used for terminating sensor leads shall be steel and painted in accordance with either of the following methods:

a. One coat of wash pretreatment in accordance with type I of TT-C-490 (0.0005 to 0.0007 inch thick).

b. A hot dip-tank phosphate treatment in accordance with type I of TT-C-490, followed by one coat of primer in accordance with TT-P-1757 (0.0002 to 0.001 inch thick).

Primer shall be top-coated with two coats (each approximately 0.001 inch thick) of gray enamel in accordance with MIL-DTL-15090.

3.3.5 Glass edge finish. Corners shall be nipped, sharp edges shall be deburred, and flares shall be ground.

3.3.6 Glass edge contour. The contour of any edge shall not, when unstressed, deviate more than 0.030 inch in a span or distance of 12 inches.

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3.3.7 Temper. Temper of strengthened glass components shall be in accordance with ASTM C1048. Per ASTM C1048, heat strengthened glass shall have a surface compression of 3,500 to 7,500 pounds per square inch (psi) and fully tempered glass shall have a minimum surface compression of 10,000 psi or an edge compression of 9,700 psi.

3.3.8 Optical and mechanical properties. Optical and mechanical properties of glass not specified herein shall be in accordance with ANSI/SAE Z26.1.

3.3.9 Original luminous transmittance. The original luminous transmittance for laminate without films when tested in accordance with ASTM D1003 using illuminant C shall be:

a. 78 percent for assemblies of overall thicknesses up to ⅝ inch,

b. 70 percent for assemblies from ⅝ to 1¼ inches, and

c. 65 percent for assemblies 1¼ to 2 inches.

For laminates greater than 2 inches or laminates that contain one or multiple films, the original luminous transmittance shall be as specified (see 6.2).

3.3.10 Resistance to delamination. When the windows are tested as specified in 4.5.1, there shall be no chipping or electrical failure, and delamination shall not be evident beyond 1 inch from the edge of the glass.

3.3.11 Film, electrical conducting. Unless otherwise specified (see 6.2), the electrical conducting film shall be embedded within the inside surface of the glass laminate nearest the surface that is to be exposed to the weather. With the voltage specified (see 6.2), the resistance value of the film shall provide a wattage power density of 350±50 watts per square foot for the window. Embedded sensing elements shall be located at control points as specified in the applicable drawing specified in 3.3.2. The current drawn by any single window shall not exceed 10.0 amperes (A), or be less than 1.50 A, for improved power density. The desired power dissipation and tolerance shall be as specified (see 6.2), with exception taken in cases where geometric aspects of a transparency yield electrical parameters that are outside the allowable ranges of tolerance.

3.3.12 Panel terminals. Panel terminals shall consist of two braided or stranded copper leads, soldered or otherwise securely attached to the internal busses.

3.3.13 Bus-bar construction. There shall be internal bus-bars of a size, shape, and material in accordance with this specification which shall afford maximum visibility. The bus-bars shall make intimate electrical contact with the film. Busses shall be inset from the edge of the panel as suggested on the applicable drawing specified in 3.3.2 or to optimize power density performance for the application.

3.3.14 Electrical isolation. When tested as specified in 4.5.3.2, results shall indicate complete electrical isolation between the sensing element and its environment.

3.4 Controllers. The controller shall be electromechanical, magnetically operated, and non-programmable for direct control of equipment. The controller shall have no external ports or programmable or remotely changeable operational characteristics. Changing operational characteristics shall require opening the controller enclosure and physically making the adjustment or modification. If required, supervisory status signals (start, stop, position, on, or off) shall be achieved via one-way non-IP communication paths only. No volatile or non-volatile memory shall be present. Wireless communication capabilities shall be disabled, and no wireless connections shall be present. To prevent glass from overheating, controllers shall be a fail-safe design comprising a second comparator circuit that shuts down the heater circuit should the window begin to overheat. Controllers shall be built to the general requirements of MIL-DTL-917. Controllers shall be in a steel dripproof enclosure in accordance with Drawings 804-2226902 and 804-4353964. Controllers shall be a solid state design. Electro-mechanical relays, thermostats, and mercury devices shall not be used. Controllers shall be free of defects as specified in 4.4.2. The mounting details of the controller shall be in accordance with Drawings 804-2226902 and 804-4353964. Controllers shall be of the “ON-OFF” type utilizing both halves of the voltage wave and shall not impose a ground on the power system to which it is connected. Solid-state relays shall be in accordance with MIL-PRF-28750. ONOFF switches shall be in accordance with MIL-PRF-8805.

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3.4.1 Type I controller. Type I controllers shall be actuated by a temperature sensor embedded in the window via the junction box (see 3.3.4). Plastic shall not be drilled to insert sensors.

3.4.2 Type II controller (for retrofit use only). Type II controllers shall be actuated by a temperature sensor contacting the inside surface of the window. Windows with a notched area shall have the temperature measured in the notched area.

3.4.2.1 Sensor housing. Sensors used with type II controllers shall be mounted in a protective housing protruding from the back of the controller. Sensor housing shall be spring loaded against the surface of the window to provide intimate thermal contact with the interior side of the glass panel.

3.4.3 Controller operation. Type I and type II controllers shall be actuated by sensors and shall be capable of handling 4.4 kilowatts (kW) maximum at 440 volts root mean square (V_{rms}). The controller shall be operable at sensing element temperatures as low as -20 °F, shall open the circuit when the temperature at the sensing element reaches 90±10 °F, and shall close the circuit at a sensing element temperature no greater than 70 °F (see 4.5.4.1, 4.5.4.2, and 4.5.4.3). The controller shall open the circuit should the following occur:

a. Open circuit in sensing element.

b. Shorted circuit in sensing element.

3.4.4 Reliability. Controllers shall be designed such that the mean cycles between failure exceeds 100,000, where failure is defined as any condition that prevents the controller from meeting any requirements herein (see 4.5.5).

3.4.5 Electromagnetic interference characteristics. The controller shall comply with emission and susceptibility limits for surface ships (above deck) in accordance with MIL-STD-461 (see 4.5.4.4).

3.4.6 Interchangeability. Like controllers, assemblies, subassemblies, and replaceable parts shall be in accordance with the interchangeability requirements of MIL-DTL-917 (see 4.5.6). In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4.7 Connectors. Ship's power to the junction box and controller shall be by means of an electrical connector in accordance with SAE AS50151. The size of connector shall be in accordance with Drawings 804-2226902 and 804-4353964. Mating connectors shall be supplied.

3.4.8 Identification plate. Each controller shall be provided with an identification plate containing the manufacturer's name, catalog number, National Stock Number (if applicable), voltage, frequency, and maximum current rating. The plate shall be located on the outside of the controller. Terminals in the controller shall be permanently and clearly marked with an identification as to the function of each terminal. The identification plate shall comply with the requirements of MIL-STD-130.

3.4.9 Painting. The controller shall be painted as specified in 3.3.4.

3.4.10 Maintenance. Ordinary maintenance in service shall not require special tools, excessively precise workmanship, or critical adjustments.

3.4.11 Prohibited and restricted materials. Controllers shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or AIA NAS411-1 for all other programs. SAE AS50151 permits the use of cadmium plating; however, cadmium is a NAVSEA-restricted chemical and shall be targeted for elimination. Where cadmium cannot be avoided, see T9070-AL-DPC-020/077-2 concerning the use of restricted chemicals and ensure compliance with OSHA Hazard Communication Standard (29 CFR 1910.1200), including the placement of warning signs and labels in visible locations, and access to safety data sheets (SDS).

3.5 Electric power. Unless otherwise specified (see 3.3.11 and 6.2), windows and their controllers shall operate on 440 V_{rms}, 60 hertz (Hz), single phase, and shall provide up to 10 A to the window heater element. The equipment shall perform as specified herein when operated with shipboard type I power in accordance with MIL-STD-1399-300-1 and MIL-STD-1399-300-2.

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3.6 Thermal shock. The window shall not crack or delaminate as a result of thermal shock when tested as specified in 4.5.2.

3.7 Power constants. Power constants for windows shall be defined as follows (see 4.4.1):

a. $Kh = \frac{\text{Power at the hot spot}}{\text{Power at the control point}}$

b. $Ka = \frac{\text{Average power}}{\text{Power at the control point}}$

c. $Km = \frac{\text{Average power}}{\text{Power at the hot spot}}$

The power constants shall be:

d. Kh equal to or less than 1.60

e. Ka equal to or greater than 0.65

f. Km equal to or greater than 0.50

Power constants shall be determined by the glass manufacturer before lamination of the glass. When specified (see 6.2), records shall be available for inspection.

3.8 Vibration. When subjected to type I environmental vibration in accordance with MIL-STD-167-1 up to and including 50 Hz, windows and controllers shall show no evidence of breakage, delamination, permanent deformation, loosening of parts, change of operational state, or loss of serviceability (see 4.5.8).

3.9 Shock. Controllers and windows shall be able to withstand grade A, class I, type A shock acceptance requirements of MIL-DTL-901 without electrical or mechanical damage including no evidence of breakage, permanent deformation, or loosening of parts (see 4.5.9). Change of operational state is allowed; however, serviceability shall be retained so that the controller can be manually set or reset to the desired operational state.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein shall be classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on windows and controllers when a first article sample is required (see 3.1). This inspection shall include the examination and tests specified in table I. Tests shall be conducted in the order shown.

TABLE I. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Windows:				
Construction and assembly	X	X	3.3, 3.3.2, 3.3.3, 3.3.4, 3.3.11, 3.3.12, 3.3.13	4.4.1
Sensor characteristics	X		3.3.1	4.5.3.1
Glass edge finish	X	X	3.3.5	4.4.1
Glass edge flatness	X	X	3.3.6	4.4.1
Temper	X	X	3.3.7	4.4.1
Optical and mechanical properties	X	X	3.3.8	4.4.1
Original luminous transmittance	X	X	3.3.9	4.4.1

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TABLE I. First article and conformance inspections – Continued.

Delamination	X		3.3.10	4.5.1
Electrical isolation	X		3.3.14	4.5.3.2
Thermal shock	X		3.6	4.5.2
Power constants	X	X	3.7	4.4.1
Vibration	X		3.8	4.5.8
Shock	X		3.9	4.5.9
Controllers:				
Construction and assembly	X	X	3.4, 3.4.1, 3.4.2, 3.4.2.1, 3.4.7, 3.4.8, 3.4.9, 3.4.10, 3.5	4.4.2
Controller operation	X		3.4.3	4.5.4.1, 4.5.4.2, 4.5.4.3
Reliability	X		3.4.4	4.5.5
Electromagnetic interference characteristics	X		3.4.5	4.5.4.4
Interchangeability	X		3.4.6	4.5.6
Prohibited and restricted materials	X		3.4.11	4.5.7
Vibration	X		3.8	4.5.8
Shock	X		3.9	4.5.9

4.3 Conformance inspection. Conformance inspection shall include the examinations specified in table I.

4.3.1 Inspection lot. An inspection lot shall consist of windows or controllers of the same type, construction, and electrical requirements produced under essentially the same conditions and offered for inspection at one time.

4.3.2 Sampling for conformance inspection. A random sample of windows and controllers shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the acceptance quality limit (AQL) specified (see 6.2). Each sample that fails to meet the requirements specified in table I shall be rejected.

4.4 Examination.

4.4.1 Windows. Windows shall be examined as specified in table I to determine conformity to the applicable requirements of ANSI/SAE Z26.1.

4.4.2 Controllers. Controllers shall be examined as specified in table I for conformance with the requirements for construction and assembly.

4.5 Tests.

4.5.1 Resistance to delamination. Windows shall be cold-soaked at a temperature not warmer than -20 °F for sufficient length of time to ensure a uniform temperature throughout the thickness of the windows. The windows may be tested in a rigid fixture. Windows shall be examined to determine conformity to the requirements of 3.3.10.

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4.5.2 Thermal shock. The window shall be mounted on a cold chamber so that the heated face is on the cold side and the opposite face is at laboratory ambient with the cold side maintained at -20 °F and held for a period of 16 hours for stabilization (see 3.6). The specified voltage shall be applied through a controller for 30 minutes and then the voltage removed for a period of 3½ hours. This shall be repeated for a total of ten cycles.

4.5.3 Electrical tests (windows).

4.5.3.1 Sensor characteristics. The resistance of the embedded sensing element shall be tested with an ohmmeter to determine conformity to its rating in accordance with 3.3.1.

4.5.3.2 Electrical isolation. Electrical isolation between the sensing element and its environment shall be tested as follows to determine conformity to the requirements of 3.3.14.

a. The conventional high potential dielectric strength test shall be performed. A potential of 1,500 V_{rms} shall be applied and maintained for 1 minute between each terminal of the sensing element and each terminal of the load. The circuitry associated with the sensing element shall be protected from the high potential during the test.

b. For non-icing windows having metal inserts or frames, the test specified in 4.5.3.2.a above shall be performed. Potential shall be applied between each sensing element terminal and the metal inserts or frames of the windows.

4.5.4 Electrical tests (controllers).

4.5.4.1 Controller operation. The controller shall be tested by mounting the window in such a way that the temperature on the outboard side of the window is low enough to cause the controller to cycle, applying the rated voltage to the controller and window assembly for a period of 24 hours (minimum 200 cycles). The temperature shall be recorded at the sensing element and shall be from 70 to 100 °F. Results of the tests shall be in accordance with 3.4.3.

4.5.4.2 Open or shorted sensing element. The fail-safe function of the controller shall be tested by opening and shorting the sensing element circuit with rated voltage applied. Results of the tests shall be in accordance with 3.4.3.

4.5.4.3 Load. Controller operation shall be demonstrated at current values of 1.5 A, 4.0 A, and 10.0 A (ten cycles at each current level) when a variable resistance load is applied across the controller in accordance with 3.4.3.

4.5.4.4 Electromagnetic interference. The controller shall be subjected to the emission and susceptibility tests for surface ships (above deck) in accordance with MIL-STD-461 and 3.4.5.

4.5.5 Reliability. Controllers shall meet the requirements for reliability in accordance with 3.4.4. The prediction procedures found in MIL-HDBK-217 may be used for guidance.

4.5.6 Interchangeability of controllers. The dimensions listed below shall be gaged or measured to determine conformity to the physical interchangeability requirement of 3.4.6:

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Location of hinges and fasteners on separable parts or assemblies that must mate, such as cases, covers, and mountings.

c. Location of connectors, locking pins, fasteners, slide and mounting that receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.5.7 Prohibited and restricted materials. Prohibited and restricted materials shall be verified as specified (see 3.4.11 and 6.2).

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4.5.8 Vibration. Windows and controllers shall be tested in accordance with MIL-STD-167-1, type I vibration to the requirements of 3.8. Type II controllers shall be tested mounted on the window. Type I vibration tests described in MIL-STD-167-1 require that equipment be tested and that resonances (see 6.5) be noted. The resonances endurances tested shall be those determined in accordance with the above. The total resonant frequency endurance test time shall be at least 2 hours at the resonant frequencies. Thus, if a total of twelve resonant frequencies, as defined herein, are noted for the three test orientations, each of the twelve frequencies shall be endurance tested for at least 10 minutes.

4.5.9 Shock. Windows and controllers shall be tested in accordance with MIL-DTL-901 to the requirements of 3.9. Type II controllers shall be tested mounted on the window.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Non-icing windows are intended for use in pilot houses or similar locations in naval ships or shore installations where constant clear vision is required under icing and fogging conditions down to -20 °F with a 40-knot wind and an ice accumulation rate of 6 inches per hour. Existing stock of windows and controllers manufactured to MIL-W-18445D dated 29 December 1980 is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of controller (see 1.2).

c. When first article inspection is required (see 3.1).

d. Material of frame, thickness of the glass, exterior dimensions of the glass, clear opening dimensions of the window, and whether the frame is fixed or hinged (see 3.3.2).

e. Original luminous transmittance for laminates greater than 2 inches or laminates that contain one or multiple films (see 3.3.9).

f. If electrical conducting film is to be applied other than on the inside surface of the glass (see 3.3.11).

g. Electric power and current rating of window if other than specified (see 3.3.11 and 3.5).

h. Amount of power dissipation and tolerance (see 3.3.11).

i. Requirement for prohibited and restricted materials verification (see 3.4.11 and 4.5.7).

j. Records of power constants determinations when required (see 3.7).

k. AQL of 1.0 (see 4.3.2).

l. Packaging requirements (see 5.1).

6.3 First article inspection. The Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired to be tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.5 Resonance definition. The conditions under which the vibration amplitude of the equipment under test is equal to or greater than 2.0 times the values denoted in the vibratory displacement of environmental vibration table in MIL-STD-167-1.

6.6 Subject term (key word) listing.

Frame

Optical

Sensor

Temper

Transmittance

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 2090-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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